Sprint #8 - Lit Review

...

USSF SSC Context-Aware Anomaly Detection

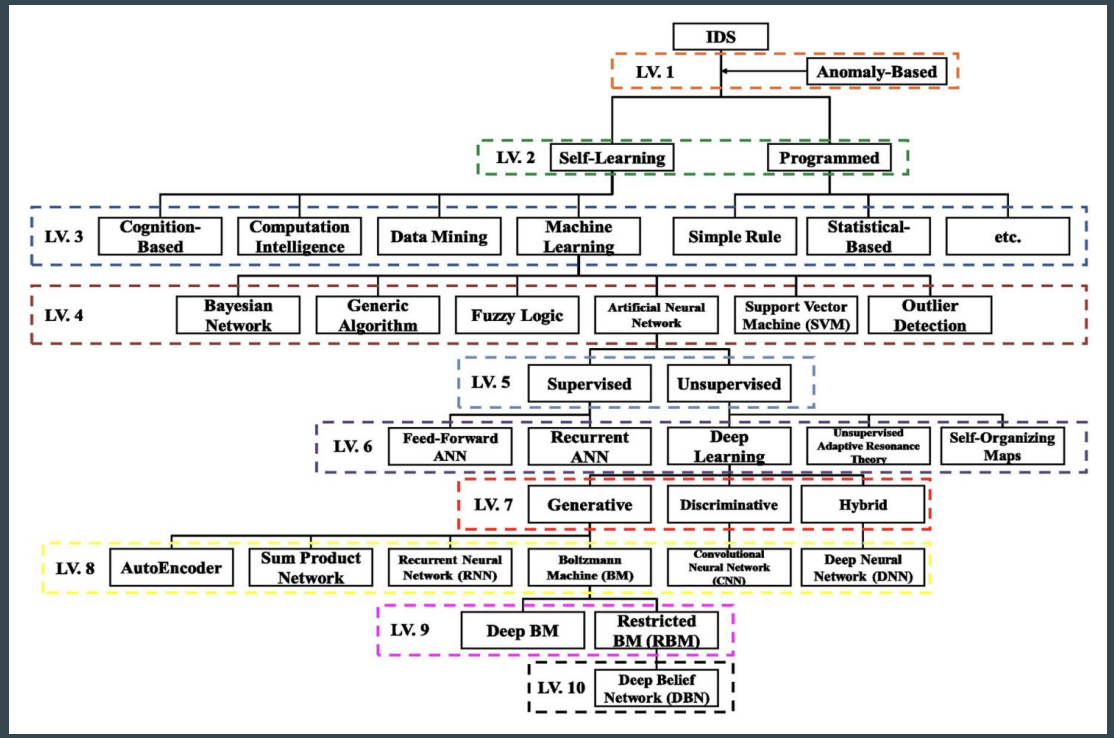
Overview

When compared to traditional anomaly detection systems, there are numerous advantages to using AI based methods in order to detect potential threats on a large network. There are many different ML models that can be used for any given task, each with different strengths and weaknesses. We would like to use a model that maximizes detection accuracy, while remaining scaleable and trainable on existing data.

Overview - Levels of Intrusion/Anomaly Detection Methods

- Lvl 1: Anomaly Based Methods
- Lvl 2: Learning Strategies
- Lvl 3: Methodologies
- Lvl 4: Classical ML Algorithms
- Lvl 5: ML Learning Strategies
- Lvl 6: Early Network Types
- Lvl 7: DL categories
- Lvl 8: Early DL Network Types
- Lvl 9: Boltzmann Machines Types
- Lvl 10: Layered RBMs

* Omitted: Reinforcement learning & newer multi-modal & graph based methodologies developed post 2017



Source: Kwon et al., A Survey of Deep Learning-Based Network Anomaly Detection, Cluster Computing, 2017.

The Role of Supervised Learning

Supervised anomaly detection relies on labeled data (normal vs attack/anomaly) to train classification models that can recognize known attack patterns with high precision and low false positive rates.

In the USSF SSC context, anomaly detection is most effective when unsupervised and supervised learning act as partners rather than competitors. Unsupervised models serve as the "explorer," analyzing raw, unlabeled data to identify "unknown unknowns" and hidden patterns without human guidance. While these models are excellent at finding outliers in large, complex datasets, they often struggle with high false positive rates and difficult to tune thresholds. Supervised learning acts as the "expert" layer that refines these discoveries, using labeled historical data to validate findings and transform raw anomaly detection into actionable intelligence.

Use Cases for Supervised Learning

Point anomalies are individual data instances that deviate significantly from the rest of the dataset, such as a singular massive spike in network traffic or a sensor reading far outside its expected range. While unsupervised models like Isolation Forests excel at isolating these outliers quickly,

supervised learning provides the critical validation layer needed to reduce false positives. By training on labeled historical "known-bad" points, supervised classifiers can instantly distinguish between a harmless technical glitch and a genuine malicious intrusion that follows a recognized pattern.

A contextual anomaly (or conditional anomaly) occurs when a data point appears normal in isolation but becomes anomalous when viewed within a specific environment or time-frame—for example, high energy usage at noon might be "normal," but the same usage at 2 AM is a threat.

Supervised learning is indispensable here because it can incorporate "behavioral attributes" and "command encoding" to teach the system the logic of the environment. As noted by NASA JPL, nearly 41% of spacecraft anomalies are contextual, and supervised models help bridge the gap by identifying these logic-defying events that unsupervised systems might overlook as simply "diverse data".

Collective anomalies represent a set of related data points that are anomalous as a group, even if each individual point appears normal when taken alone. A primary example is a slow and low DDoS attack, where individual requests are indistinguishable from normal traffic, but their collective volume reveals a coordinated strike.

Supervised learning assists by training on sequence-based labels, allowing models like LSTMs to recognize the specific "rhythm" of a systemic failure or a multi-stage attack. This helps turn the "reconstruction error" of unsupervised deep learning into a clear, labeled alert for a sustained or evolving threat.

Unsupervised Learning - An Overview (by Kristian, Sneha)

What is it?

- Unsupervised Learning is a ML approach that looks for patterns in data sets without the structure of labeled data and targeted outcomes.
- “Unsupervised ML is widely used to mine large, unlabeled datasets to make data-driven discoveries in critical domains” (Chang et al., 2025)

Why use it for anomaly detection?

- Since you won’t know the anomaly without establishing the baseline, unsupervised ML works by building a model of what normal looks like, and classification any deviation as a possible anomaly .
- “Examples of major unsupervised tasks include...anomaly detection to discover informative outliers” (Chang et al., 2025)

Source: https://arxiv.org/html/2506.04553v1

Unsupervised Learning - Challenges and Validation Techniques

- "By nature, unsupervised learning problems do not include a target outcome...complicating the the tasks of model comparison and hyperparameter selection" (Chang et al., 2025)
- Supervised learning usually has a target outcome to test model reliability. For example, in image classification each image has a "correct" answer: an image of a bulldog mapped to the word dog, and image of a Ford matched to the word car.
- Unsupervised learning doesn't have this target outcome. There is no way to know if the model is correct in a way, which complicates the task of comparing model effectiveness and determining which parameters to tweak.

- Although you can't match the output of a unsupervised model to one correct answer, there are validation techniques that can be applied to resultant data.
- "Applying multiple techniques can help facilitate validation...the trustworthiness of a finding can be measured by the frequency by which it appears across different modeling results" (Chang et al., 2025)
- "Validation techniques are needed to assess reliability...unlike supervised learning, there is no ground-truth target to compare our findings against" (Chang et al., 2025)
- Some validation techniques include model agreement, human review, and synthetic anomaly detection.

Source: https://arxiv.org/html/2506.04553v1

Unsupervised Learning Models for Anomaly Detection - Isabella

Name	Isolation Forests (IF)	One-Class Support Vector Machine (OC-SVM)	Local Outlier Factor (LOF)
Description	A decision tree based model that finds anomalies through tree splits, since these usually have fewer tree splits to isolate than normal data.	Learns what “normal” data appears as and creates a boundary around that. Anything outside the boundary is an anomaly.	Compares point clusters, looking for points that are farther away and marking them as anomalies.
Pros	Works great on large/complex datasets. Works well with numerical data.	Works well even if there are very few or no anomalies in data.	Great at finding points unusual to local surroundings, automatically adjusts to local densities/irregular data.
Cons	Smaller anomalies may be missed.	Sensitive to kernel type, slow and computationally heavy for large data sets.	Has a harder time scaling to larger datasets.

Sources

Reference 1 - Anomaly detection using unsupervised machine learning algorithms: A simulation study

Description: This study covers an evaluation of five different machine learning algorithms, including isolation forests, one class SVM models, and Local Outlier Factor models. The study shows that *Isolation Forests outperformed* the other models in its ability to juggle both precision and recall.

Reference 2 - Robust IoT security using isolation forest and one class SVM algorithms

Description: This article delves into anomaly detection in internet networks using isolation forest and one class SVM models.

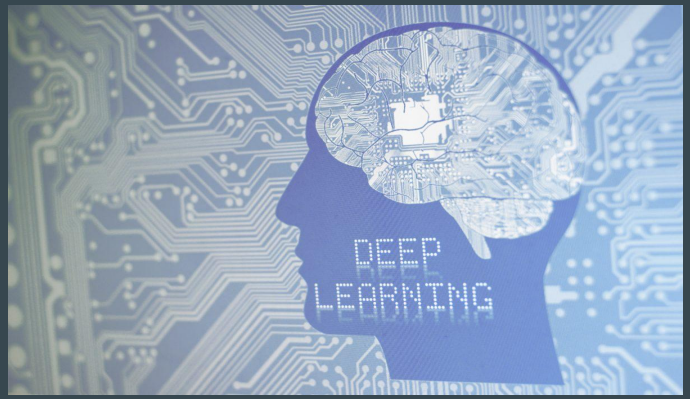
Reference 3 - On the evaluation of outlier detection and one-class classification: a comparative study of algorithms, model selection, and ensembles

Description: This compares the one class SVM model with other outlier detectors. It also covers model selection and evaluates the performance of various models.

Deep Learning- Ishaan Buddharaju, Jose Zamora, Travis Coleman

What is Deep Learning?

Deep learning is a form of machine learning the focuses on artificial neural networks. It uses a multi layer network meaning it builds on itself to generate stronger outcomes with every input and output. These models are very versatile because of their ability to use both structured and unstructured data sets. Unlike other forms of machine learning, with a consistent flow of data and parameters, deep learning models never plateau. For our use case of anomaly detection there are many methods that can be applied to these models.



Deep Learning- Methods for Anomaly Detection

-Travis C.

Methods based on **reconstruction** - Reconstruction based approaches revolve around training a model on the normal distribution of data, and once trained, the model tries to reconstruct the data, and the reconstruction error (the difference between the original data and its reconstruction) is used as an indicator for an anomaly.

Methods based on **predictions** - Prediction based approaches train a model to predict future data points and instead of focusing on reconstruction (how well a model can rebuild an output) they detect anomalies by comparing predicted values with actual observed values, large prediction errors are indicated as an anomaly.

Hybrid Methods- These approaches combine traditional anomaly detection methods with deep learning models to use the strengths of both. Combining both methods allows hybrid models to have better interpretability, and accuracy while also being able to handle complex high dimensional data.

Lit Review- Deep Learning Anomaly Detection Methods -Travis C.

“Deep Learning Advancements in Anomaly Detection: A Comprehensive Survey”
Authors: Haoqi Huang, Ping Wang , Jianhua Pei , Jiacheng Wang Shahen Alexanian, and Dusit Niyato
Relevance: This paper directly explores the applications and methods of using deep learning models to detect anomalies.
Findings: ● Deep Learning Methods are effective for Anomaly Detection ● Reconstruction based methods are useful for unsupervised settings but can struggle when anomalies closely resemble normal data ● Prediction based methods are useful for time-series and sequential data since they detect temporal patterns and can detect unexpected changes ● Hybrid approaches are great since combine deep learning with traditional anomaly detection methods in order to improve interpretability and accuracy

Deep Learning - Conditional Variational Autoencoders

Jose Z

CVAE is a deep learning model that does not focus on one right answer but rather generates many possible “right” answers. They are commonly used for “hallucinating” possible outcomes and failure states using conditional data. In anomaly detection It can be used for predicting and generating new synthetic data for potential unknown anomalies that we may not have foreseen.

Pros	Cons
Data scarcity- very useful when large amounts of failure data is not available	**Consumption**- Requires a lot of computing power compared to other learning models.
Context aware- Has an understanding of normal data within the defined context.	**Accuracy**- Not all the synthetic data generated is usable or is too noisy.

Deep Learning -Time Series Learning

- Brian S.

What are Time Series Models?

Algorithms that analyze data collected sequentially over time, using the chronological order of observations to capture patterns between past and future values. Unlike traditional models that treat data points independently, time series models leverage time as the independent variable to identify trends, seasonality, and anomalies, and to forecast future values based on previous observations in the sequence.

Statistical Methods:	Deep Learning Methods:	Distance-Based Methods:
- ARIMA/SARIMA - Exponential Smoothing	- LSTM (Long Short-Term Memory) - CNN (Convolutional Neural Networks) - GRU (Gated Recurrent Units)	- Matrix Profile

Deep Learning - Time Series Learning (ARIMA/SARIMA) - Brian S.

ARIMA (Autoregressive Integrated Moving Average) is a statistical forecasting method that predicts future values based on past observations. It combines three components: AR (autoregressive - uses past values), I (integrated - removes trends by differencing the data), and MA (moving average – adjusts using past errors).

SARIMA extends this by adding seasonal components to capture repeating patterns at fixed intervals.

PROS	CONS
- Interpretable: Clear statistical foundations, explainable results - Efficient: Fast training/inference, works with small datasets - Well-established: Strong theoretical foundation, proven track record - Simple configuration: Fewer parameters than deep learning	- Linear only: Cannot capture complex non-linear patterns - Requires stationarity: Needs data preprocessing/transformation - Manual tuning: Expert knowledge needed for parameter selection - Single-series focus: Doesn't scale well to multi-sensor systems

Time Series Learning (ARIMA/SARIMA) - Literature Review - Brian S.

"An Experimental Review on Deep Learning Architectures for Time Series Forecasting"	"A Novel Anomaly Detection Algorithm for Hybrid Production Systems based on Deep Learning and Timed Automata"
Authors: Lara-Benítez et al. (2021)	**Authors:** Hranisavljevic et al. (2016)
Relevance: • Direct ARIMA vs. deep learning comparison across diverse domains • Demonstrates ARIMA's limitations with non-linear patterns common in complex telemetry systems	**Relevance:** • Addresses hybrid systems applicable to spacecraft telemetry • Motivates why single-modality approaches like ARIMA are insufficient for multi-source data fusion
Findings: • ARIMA outperformed by LSTM/CNN • Requires more expertise/preprocessing • 38,000+ models tested • Insufficient for complex dynamics	**Findings:** • ARIMA separates signal types, limiting analysis • Need integrated approaches for timing dependencies • Tested on systems similar to space applications • Shows statistical → deep learning evolution

Time Series Learning (LSTM Autoencoders)

- Brian S.

LSTM (Long Short-Term Memory) Autoencoders encode input time-series into a compressed representation, then reconstruct the original sequence. Trained on normal data, they detect anomalies through reconstruction error—sequences that cannot be reconstructed well are anomalous.

PROS	CONS
- Captures complex patterns: Handles non-linear temporal dependencies in multivariate data - No preprocessing: Works without stationarity requirements or data transformation - Multivariate: Processes multiple sensors simultaneously - Flexible: Handles both predictable and unpredictable sequences	- Data hungry: Requires substantial training data - Computationally expensive: Slow training/inference - Black box: Difficult to interpret learned patterns - Threshold selection: Requires validation data or expertise

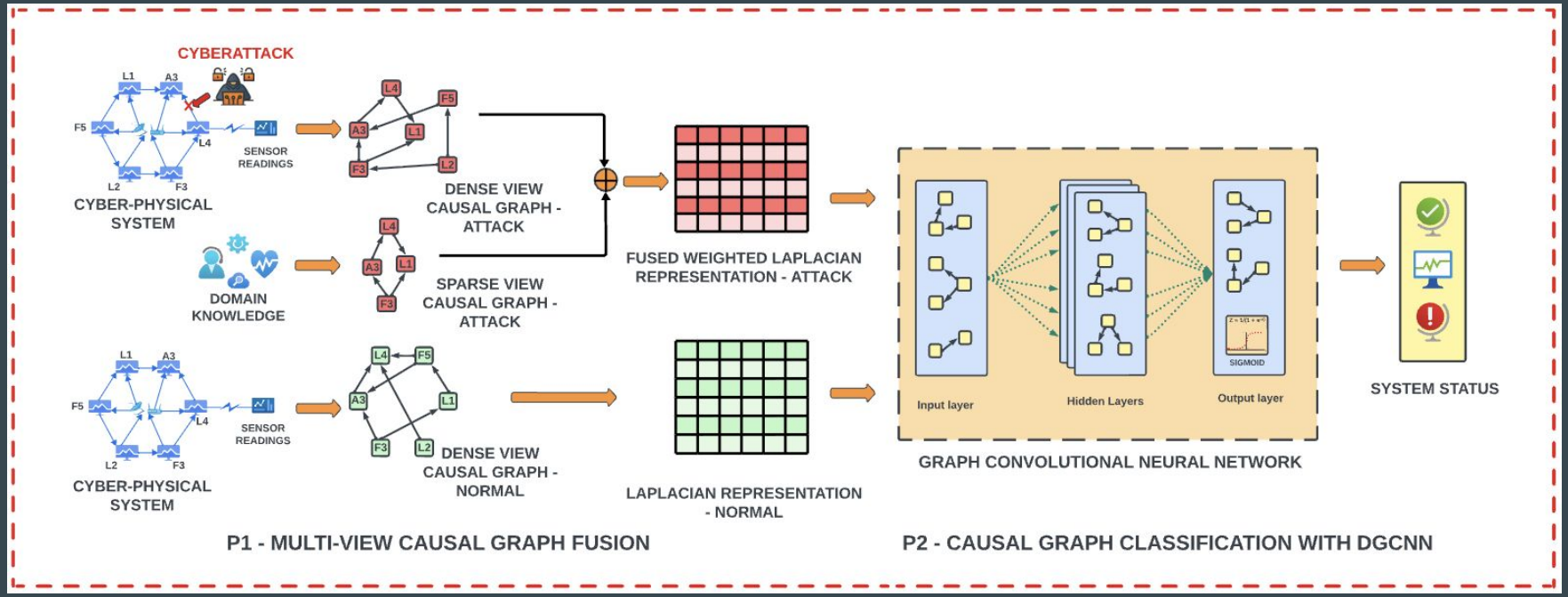
Time Series Learning (LSTM AE) - Literature Review

- Brian S.

"Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding"	"LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection"
Authors: Hundman et al. (NASA JPL) (2018)	**Authors:** Malhotra et al. (2016)
Relevance: • Real spacecraft telemetry from SMAP and Mars Curiosity • Addresses deployment challenges (false positives, production systems)	**Relevance:** • Handles unpredictable time-series (manual controls, varying loads) • Multi-sensor system focus
Findings: • 80% recall, 87.5% precision on spacecraft data • Command encoding improves predictions • Non-parametric thresholding outperformed Gaussian methods • 41% of anomalies were contextual—requiring LSTM over simpler methods	**Findings:** • Detects anomalies in unpredictable data where prediction methods fail • Works on sequences 30-500 timesteps • Outperformed prediction-based LSTM (F0.05 = 0.83 vs 0.03) • Effective across periodic, aperiodic, quasi-periodic data

Deep Learning - Causal Graph Based Fusion Methods

- Ishaan B.



Source : Malarkkan et al., Multi-View Causal Graph Fusion Based Anomaly Detection in Cyber-Physical Infrastructures (2024).

Deep Learning - Causal Graph Based Fusion Methods

- Ishaan B.

Causal graph–based anomaly detection method diverges from the correlational methods

- Each data modality (dense telemetry, sparse logs, etc.) is converted into its own causal graph Laplacian, capturing mode-specific relationships.
- These Laplacians are combined using a spectral fusion operator, producing a fused eigenspace that encodes cross-modal causal structure.
- The fused graph is then classified using a Deep Graph CNN (DGCNN) to distinguish normal vs. anomalous system behavior.

Source : Malarkkan et al., Multi-View Causal Graph Fusion Based Anomaly Detection in Cyber-Physical Infrastructures (2024).

Reinforcement Learning - RL

-Angeira PM.

Overview:

- Mistake driven learning method
  - Uses trial and error to try all possible state-action pairs to determining strategy with best long-term return
  - Uses concept of reward to judge response to environment

Pros	Cons
Does not require data labelling	Too many features can define state and reward; volatile
Errors can be corrected; realizes mistakes from rewards, avoids actions under specific environments	High resource consumption because it will try all state-action pairs
Applicable to complicated problems that require best results after series of operations, rather than single action	Agent may find loopholes instead of desired behavior

Semi-Supervised Learning - SSL (Hybrid)

-Angeira PM.

Overview:

- Bridges supervised and unsupervised methods to create more accurate, efficient, and scalable models
- Addresses scarcity of labeled data by using an initial model to assign pseudo-labels to unlabelled data
- Incorporates most common prediction into training set based on predefined assumptions (smoothness and clustering)

Pros	Cons
More applicable in the real world; labelling of limited size of data	Incorrect classification of unlabelled data could mislead model to false prediction
Relieves lack of of labelled data and ensures model has adequate training before implementation; improves accuracy	Model performance depends heavily on confidence threshold
Efficient with unstructured data	More complex training than traditional supervised learning

Applicability of RL & SSL in Anomaly Detection

-Angeira PM.

Reinforcement Learning

- Applicable Model: Deep Q-network (Q-learning & Deep Neural Networks)
- Methodology: Run multiple agents under same network to identify states and reward RL model from its prediction (more stable and robust)

Semi-Supervised Learning

- Applicable Model: K-Means (Centroid-Based Clustering)
- Methodology: Cluster data and find the center of each cluster, assign each point to its nearest centroid, then update centroids to the mean of their assigned points

RL & SSL Sources

-Angeira PM.

“Machine Learning in Network Anomaly Detection: A Survey”	“Machine Learning for Anomaly Detection: A Systematic Review”
Authors: Song Wang, Juan Fernando Balarezo, Sithamparanathan Kandeepan, Akaram Al-Hourani, Karina Gomez Chavez, Benjamin Rubinstein	Authors: Manar Abu Talib, Qassim Nasir, Fatima Mohamad Dakalbab
Relevance: - Implementation of ML models in anomaly detection under different network contexts	Relevance: - Introduction to ML algorithms - Precision comparison of each technique - Advantages and disadvantages of each technique
Findings: - Model is picked based on data set - Proportion of labeled data is a key factor in model selection	Findings: - PCA and CFS are the most used among selection/extraction techniques - Most utilized approach for anomaly detection was unsupervised (based off of 2021 study)